Annex V. Debt Sustainability Analysis Update

Approved By

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The Update on Debt Sustainability Analysis has been prepared jointly by IMF and International Development Association staff using the debt sustainability framework for low-income countries approved by the Boards of both institutions.

Risk of external debt distress:	**Moderate**
Augmented by significant risks stemming from domestic public and/or private external debt?	**Yes**

Malawi faces a moderate risk of debt distress based on an assessment of public external debt but a heightened overall risk of debt distress, reflecting significant vulnerabilities related to domestic debt.[1]Malawi's external debt situation is broadly unchanged from the last DSA, but the debt level and interest expense remains high.[2] All baseline external debt burden indicators remain below their indicative thresholds, but public external debt remains vulnerable to exogenous shocks, notably shocks to export revenues and the exchange rate. The projected borrowing path and debt policies remain broadly unchanged since the last DSA, but close attention will be needed on the financing terms of any proposed infrastructure investments given the limited headroom for further borrowing. To enhance resilience to shocks, efforts should be stepped up to further diversify the economy – particularly exports, broaden the revenue base, strengthen public finance management, and build fiscal buffers.

[1] This report updates the DSA analysis contained in IMF Country Report No. 17/183 (July 2017). The fiscal year for Malawi runs from July to June; however, this DSA is prepared on a calendar year basis.

[2] Malawi's policies and institutions are classified as "weak" under the World Bank's Country Policy and Institutional Assessment (CPIA) Index (average score in 2013–16: 3.2). The relevant indicative thresholds for this category are: 30 percent for the NPV of debt-to-GDP ratio, 100 percent for the NPV of debt-to-exports ratio, 200 percent for the NPV of debt-to-revenue ratio, 15 percent for the debt service-to-exports ratio, and 18 percent for the debt service-to-revenue ratio. These thresholds are applicable to public and publicly guaranteed external debt.

A. Background

1. **Real GDP growth of the Malawian economy picked up last year**—estimated at 4 percent, up from 2.3 percent in 2016. This mainly reflects a recovery in agricultural production following two years of drought. Inflation declined to 7.8 percent in February (after peaking at 24 percent at end-2015) due to low and relatively stable food prices, prudent fiscal and monetary policies, and a stable exchange rate. The overall fiscal balance improved by 1.7 percentage points of GDP in FY 16/17 relative to the previous fiscal year, reflecting mainly increased revenue collection. The current account deficit narrowed—to an estimated 10 percent of GDP, from 13.6 percent in 2016. The improvement reflects mainly a large base effect in 2016 from increased imports of maize and related services, which has more-than offset the impact from deterioration in the terms of trade. Real GDP growth is projected to slow to 3.5 percent in 2018 (due to the adverse effects of dry spells and armyworm infestations on agricultural production), before picking up again (to 4.5 percent) next year and stabilizing at 6 percent over the medium term.

2. **Malawi's public and publicly guaranteed (PPG) external debt stood at about US$2 billion (32.6 percent of GDP) in 2017**, up from $1.79 billion (33.2 percent of GDP) in 2016.[3] The increase in PPG external debt during 2017 mainly reflects $382 million of new disbursements (with $353 million from multilaterals and $29 million from bilateral creditors) and repayment of about US$150 million of principal.

3. **Public external debt is held mainly by multilateral creditors** (78 percent of total, Text Table 1). The main provider of loans to Malawi is the International Development Association (IDA) (43 percent), followed by the African Development Fund (ADF) (14 percent) and the IMF (11 percent). China and India are the main holders among bilateral creditors, with China accounting for about 12 percent of total debt—surpassing the IMF as the third largest creditor to the country. Data on private external debt remains unavailable, but the amounts are not believed to be large.

4. **Public domestic debt is held by the Reserve Bank of Malawi (RBM, the largest creditor), commercial banks, and the non-bank financial sector**. Gross domestic debt increased

[3] Public and PPG external debt covers that contracted and guaranteed with nonresidents by central government and the Reserve Bank of Malawi. External arrears remain zero.

from MK206.6 billion (13.8 percent of the recently rebased GDP[4]) at the end of 2012 to MK1,015 billion (22.6 percent of GDP) at the end-2017.[5] The spike reflects a progressive shift of debt from external to domestic borrowing in recent years (Text Table 2, Text Figure 1). As discussed in the last DSA, the increase largely reflects i) the rise in government net domestic financing during FY 13/14–14/15, following the drop in external financing in the wake of the cashgate scandal; ii) the securitization of domestic arrears in 2013 and 2015–16; and iii) the issuance of a substantial amount of Treasury notes over 2012–15 for the conversion of ways and means (including promissory notes in 2013–14 by the government to recapitalize the RBM following losses arising from the 2012 exchange rate devaluation) and maturing Treasury bills into longer-term government securities. For more detailed discussion of evolvement of Malawi's public and external debt and debt services in recent years, see last DSA contained in IMF Country Report No. 17/183.

Text Table 1: Composition of Public and Publicly Guaranteed External Debt
(Million U.S. dollars)

	2015		2016		2017	
	Actual	Share	Actual	Share	Actual	Share
Multilateral	**1,343.1**	**75**	**1,362.3**	**76**	**1,584.3**	**78**
IMF	162.8	9	206.1	12	223.8	11
IDA	589.9	33	642.2	36	860.1	43
ADF	228.8	13	247.9	14	290.3	14
IFAD	71.8	4	72.5	4	77.5	4
other multilateral & PTA[1]	289.8	16	193.7	11	132.6	7
Bilateral	**439.5**	**25**	**426.2**	**24**	**436.7**	**22**
Belgium	1.7	0	1.7	0	1.7	0
China	242.7	14	226.9	13	237.9	12
India	151.7	9	147.3	8	142.1	7
Others	43.3	2	50.4	3	55.0	3
Commercial	0.39	0	0	0	0	0
Total	**1,783**	**100**	**1,789**	**100**	**2,021**	**100**

Sources: Malawian authorities and IMF staff calculations.

[1] PTA is a regional non-resident bank. The outstanding amount to PTA was zero, as of end-2017.

[4] The National Statistical Office of Malawi rebased their nominal GDP series from calendar year 2007 to 2010. The 2010-based national supply and use table has been used to derive a new GDP series, which was revised upwards by 29 percent. Some key indicators have changed as a result. For instance, the revenue to GDP ratio in FY 2010/11 declined from 24.5 percent to 15.6 percent and the public debt-to-GDP in 2011 fell by 12 percentage points to 28 percent of GDP.

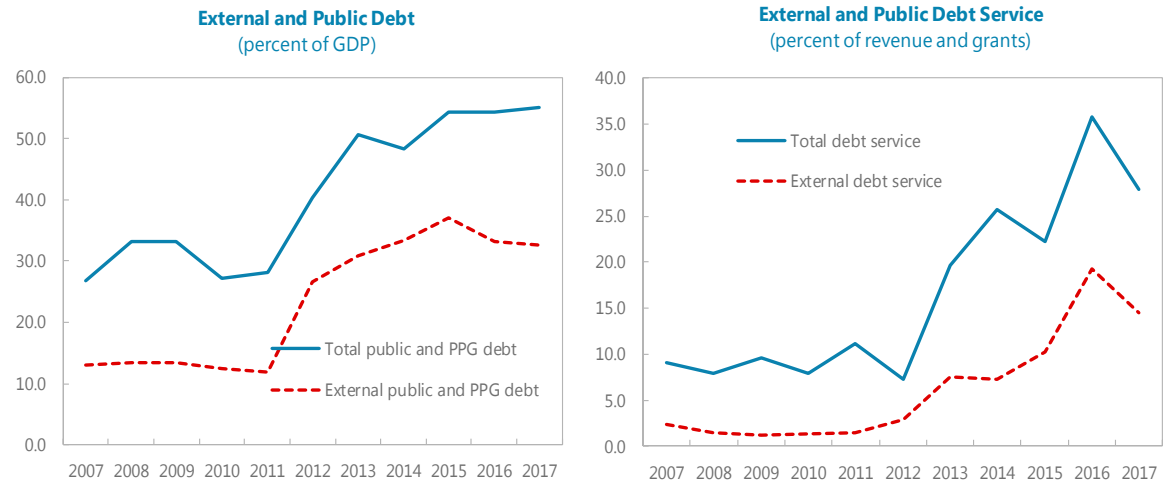
[5] Domestic debt mainly includes the total stock of treasury bills, treasury notes, local registry stocks, and ways and means advances. It also covers zero coupon promissory notes which have not yet matured, promissory notes related to the recapitalization of RBM (until end-September 2017), promissory notes for Malawi Saving Bank's toxic assets, and certified arrears for which promissory notes have not yet been issued. No contingent liabilities are included in the analysis due to a lack of data.

Text Table 2. Composition of Gross Domestic Debt (Percent of GDP)

	2012	2013	2014	2015	2016	2017
			Actual			
Treasury bills at cost value	9.0	9.1	6.9	6.2	5.8	5.6
Treasury notes	2.7	1.8	1.3	6.0	10.7	14.4
Local registered stocks (LRS)	0.2	0.1	0.0	0.0	0.0	0.0
Ways and means advances from RBM	1.7	5.2	3.0	0.9	0.9	1.0
Promissory notes	0.1	3.6	3.7	2.6	0.8	0.1
Commercial bank advances	0.1	0.0	0.0	0.0	0.0	0.0
Total	**13.8**	**19.8**	**14.9**	**15.8**	**18.2**	**21.1**

Sources: Malawian authorities; IMF staff estimates.

Text Figure 1. Public and PPG Debt



Sources: Malawian authorities
and IMF calculations.

B. Underlying Assumptions

5. **The medium- and long-term macroeconomic framework underlying the DSA is consistent with the scenario presented in the Staff Report for the 2018 Article IV Consultation and Request for a three-year arrangement under the Extended Credit Facility**. The main projections for key macroeconomic variables are summarized in Box 1 and text Table 3. The main differences between the current assumptions and those underlying the last DSA in July 2017 are:

- *GDP growth* is estimated to be lower in 2017–18 (emanating from a weaker growth in manufacturing constrained by shortages of energy and water supply, and depressed purchasing power with low agricultural commodity prices in 2017; as well as the adverse effects of dry spells and armyworm infestations on agricultural production in 2018) but higher by about 0.5–1 percentage points over the medium to long term (reflecting improved confidence, more robust agricultural production, enhanced electricity generation, improved access to finance, and implementation of growth-friendly structural reforms under the support of the ECF arrangement. All of these are expected to help unleash higher potential growth than is assumed in the last DSA);

- *A higher fiscal primary deficit* is assumed for 2017–18 due to higher primary expenditure, reflecting increased spending on wages and salaries, goods and services, maize purchases, subsidies for bailout of ADMARC, arrears payments, and domestic and foreign-financed capital spending; and

- *A larger current account deficit* throughout the period, reflecting worsened terms of trade and higher investment-related imports on the back of stronger investment under the new ECF arrangement along with stronger export growth.

Text Table 3. Macroeconomic Forecast and Assumptions (Previous and Current DSAs)

	Real GDP growth (percent)		Primary deficit (percent of GDP)		Change in public debt (percent of GDP)		Current account (percent of GDP)		FDI (percent of GDP)	
Year	Previous	Current	Previous	Current	Previous	Current	Previous	Current	Previous	Current
2014	5.7	5.7	0.2	0.2	-2.6	-2.4	-8.4	-8.3	4.9	4.9
2015	2.9	2.9	2.1	2.1	6.6	6.2	-9.4	-9.4	1.8	1.8
2016	2.3	2.3	1.9	2.1	-0.3	0	-13.5	-13.6	3.1	3.1
2017	4.5	4.0	1.3	2.9	-4.4	0.7	-9.3	-10	3	2
2018	5.0	3.5	-0.6	-0.4	-1.0	3.6	-8.2	-8.9	2.5	2.2
2019	5.5	4.5	-0.1	0.9	-1.7	-0.3	-7.8	-8.1	2.4	2.3
2020	5.5	5.0	-0.1	-0.3	-2.4	-2.1	-7.6	-7.9	2.4	2.4
2021	5.5	5.5	-0.1	-0.1	-1.5	-2.4	-7.4	-7.7	2.3	2.5
2022	5.5	6.0	-0.4	-0.3	-1.4	-2.5	-7	-7.7	2.3	2.5
2023		6.5		-0.3		-2.7		-7.6		2.5
Avg 2023-37	5.5	6.5	0.4	0.6	-0.6	-1.3	-6.2	-6.5	2.1	2.5

Sources: Malawian authorities and IMF staff calculations and projections.

Box 1. Baseline Macroeconomic Assumptions

Real GDP growth is projected to moderate to 3.5 percent in 2018 and gradually pick up and stabilize at 6.5 percent over the medium to long term. The long-term growth rate is about 1.5 percentage point higher than the average during 2008-17.

Inflation (end-of period) is projected to rise temporarily from 7.1 percent at end-2017 to 9 percent this year (reflecting the impact of a rise in maize prices from a historically low level) before resuming a gradual disinflation path. Tight fiscal and monetary policies are expected to continue anchoring inflation expectations. Inflation is projected to reach 5 percent by 2022.

Private sector credit growth is projected at 4 percent this year before strengthening to 8 percent in 2019 and averaging about 15 percent year-on-year during 2020–2023 in line with stronger real growth, compared to an 11.5 percent average envisaged in the last DSA.

The exchange rate is projected to remain constant in real effective terms.

The tax revenue to GDP ratio is expected to edge down by 0.3 percentage points in FY 17/18 from the previous fiscal year and return to a higher level of 17.8 percent by FY 22/23. In the long run, tax revenue is projected to rise gradually to around 18.5 percent of GDP in FY 36/37, thanks to progressive reforms to tax administration and policy. The tax revenue to GDP ratio is about 0.3 percentage point lower than assumed at the time of the last DSA. However, total revenue to GDP ratio remains broadly unchanged from the last DSA.

External debt will be mainly contracted with multilateral creditors on concessional terms, with the remainder being bilateral on broadly similar terms. Budget support from multilateral and bilateral donors is assumed to remain subdued in the medium term. For FY 18/19, the framework assumes about $80 million budget support grants from the World Bank.

The current account deficit is projected to shrink gradually, reflecting large and persistent import needs.

New disbursements on external loans. For 2018, new disbursements on external loans are taken from the authorities' fiscal framework, which projects capital spending covered by external loans to reach 4.8 percent of GDP in FY 17/18 and 4.1 percent in FY 18/19. It is assumed that external project loans will remain close to 2.5–2.6 percent of GDP in subsequent fiscal years.

Net domestic financing. It is assumed that government net domestic financing will be limited to less than 1 percent of GDP in each fiscal year beyond FY 18/19, contributing marginally to the change in domestic debt.

C. Debt Sustainability Analysis

External DSA

6. **Based on the assumptions outlined above, Malawi's debt is assessed to be sustainable with a moderate risk of debt distress** (Figure 1 and Table 1–3). As in the last DSA, Malawi is classified as a "weak" performer, based on the quality of the country's policies and institutions as measured by the three-year average of the rating under the World Bank's Country Policy and Institutional Assessment (CPIA). This is reflected in lower (less accommodative) debt sustainability thresholds compared to countries operating in a strong policy environment.

7. **Under the baseline scenario, all debt burden indicators remain below their indicative thresholds.** The PPG external debt is projected to peak at 32.2 percent of GDP in 2019–20, with a PV of debt-to-GDP ratio of 20 percent. The PV of PPG external debt is expected to decline gradually to 13 percent over the long term. Debt service-to-exports ratio is expected to remain high at 7.5 percent in 2018 (compared to a historical average of 4 percent), but has declined significantly from a 10-year high of 12 percent in 2016 (the latter reflects a large amortization related to the debt restructuring operation with the PTA bank which was completed in 2017). The debt service is expected to gradually decline below 4 percent of exports over the long term.

8. **A weaker debt outcome is envisaged under all but one stress test.** The strongest impact on the indicators arises under two bound tests (B3 and B6 in Table 2): i) US dollar GDP deflator at historical average minus one standard deviation in 2019–20 and ii) one-time depreciation of 30 percent in 2019. Both shocks cause the PV of debt to GDP, PV of debt to exports, and PV of debt to revenue to breach the thresholds and remain at elevated levels. Although liquidity indicators – debt services to exports/revenue ratios – do not breach the thresholds, they are expected to steepen significantly, especially under the depreciation shock for debt services to revenue ratio. Temporary shocks to real GDP growth, export growth, non-debt creating inflows such as FDI, or a combination of them (B1, B2, B4, and B5 in Table 2) are also expected to cause the debt path to breach most thresholds for multiple years. Under the historical scenario (A1 in Table 2), when the average current account deficit was around 10.1 percent of GDP[6] and FDI was low (around 1.5 percent of GDP), all ratios are projected to breach the thresholds and remain at elevated levels. However, the probability

[6] Based on BPM6. Project and dedicated grants were classified on the current account but under the BPM6 are classified on the capital account leading to significant increase in historical values of current account balances. For example, the current account deficit averaged 10.1 percent over last 10 years under the revised classification, compared to 5.5 percent previously. Current account deficit for 2011 and 2012 were 9.3 and 8.7 percent, respectively, up from 5.9 and 3.5 percent previously.

is low for Malawi to repeat history by running high and protracted current account deficits in the medium to long term. Following a devaluation in 2012, Malawi adopted a flexible exchange rate regime and the exchange rate is now expected to serve as a shock absorber. Moreover, FDI inflows are expected to pick up steadily as macroeconomic stability is regained and the business environment improves. These flows are expected to concentrate on the agriculture, manufacturing, and energy sectors, helping boost export growth and reduce the heavy reliance on energy-related imports (such as generators).

Public DSA

9. **The risk of debt distress is heightened significantly when adding domestic public debt to external debt** (Figure 2, Table 4–5). The total public debt is projected to peak at 54.4 percent in 2019 before receding as the fiscal primary balance starts to turn positive. The peak of debt is nearly 5 percentage points higher than projected in the last DSA, reflecting a higher primary deficit and a higher average real interest rate (partly reflecting lower inflation) on domestic debt despite stronger real GDP growth. Similarly, total debt service is expected to remain above 10 percent of revenue and grants over the medium term. This points to the importance of obtaining the most favorable financing terms going forward. In PV terms, total public debt (at 43 percent of GDP in 2018) is projected to breach the LIC DSA benchmark of 38 percent (for those countries with weak policies and institutions) and remain above the benchmark until 2023. This reflects a worsening of the primary deficit compared to the last DSA, notably in 2017 and 2018 (Text Table 3).

10. **Standard tests illustrate the importance of the primary deficit in determining the debt trajectory**. Under the scenario (A1 in Table 5) with real GDP growth (of 5 percent) and primary balance at historical averages (of 1.3 percent), the PV of total public debt will be slightly worse than the baseline, with the PV debt-to-GDP ratio above 38 percent before 2023. The most severe impact comes from a temporary shock of a one-time 30 percent real depreciation in 2019, which will significantly worsen the debt path, sending the peak PV of total public debt to 51 percent of GDP in 2019.

11. **The gross domestic debt to GDP ratio is projected to decline gradually over the medium term**, to around 15 percent of GDP at end-2023. These projections assume that:
i) the cost value of all maturing Treasury bills and the face-value of all maturing Treasury notes will be continuously rolled over; ii) government's net domestic financing will be limited to less than one percent of GDP beyond FY 18/19; and iii) all maturing promissory notes will be automatically converted into advances from the central Bank and ultimately rolled over into Treasury bills of

varying maturities (91, 182 and 364 days) or Treasury notes of longer than one year maturity which could then be tradable in the secondary market.

D. Conclusions

12. **Malawi remains at moderate risk of debt distress, based on an assessment of external public debt, but overall risks have heightened compared to the last DSA, reflecting rising vulnerabilities to domestic debt and external conditions.** Malawi is historically vulnerable to shocks, particularly those related to weather, commodity prices, and exports, which often take on many dimensions at the same time threatening macroeconomic stability. Absorption of these shocks while maintaining macroeconomic stability and debt sustainability will require careful macroeconomic management and difficult policy choices.

- *Prudent project selection*. Given limited headroom for further borrowing, close attention and prudence should be applied to project identification and the financing terms of any proposed infrastructure investments while balancing needs with rates of return and debt sustainability concerns.[7]

- *Broadening revenue base and strengthening public financial management*. Given Malawi's high aid dependency, risks of negative financing shocks in the form of delayed donor support, or lower-than-expected revenue collection remain. Such an environment requires further efforts to maximize the impact of finite domestic resources, including broadening the tax base and strengthening public procurement and financial management.

- *Building fiscal buffers*. Large domestically-financed fiscal deficits in recent years means that Malawi is reaching the upper limits at which its stock of domestic debt can be sustainably managed. Measures to increase fiscal buffers to enable the budget to withstand the impact of future external shocks are important.

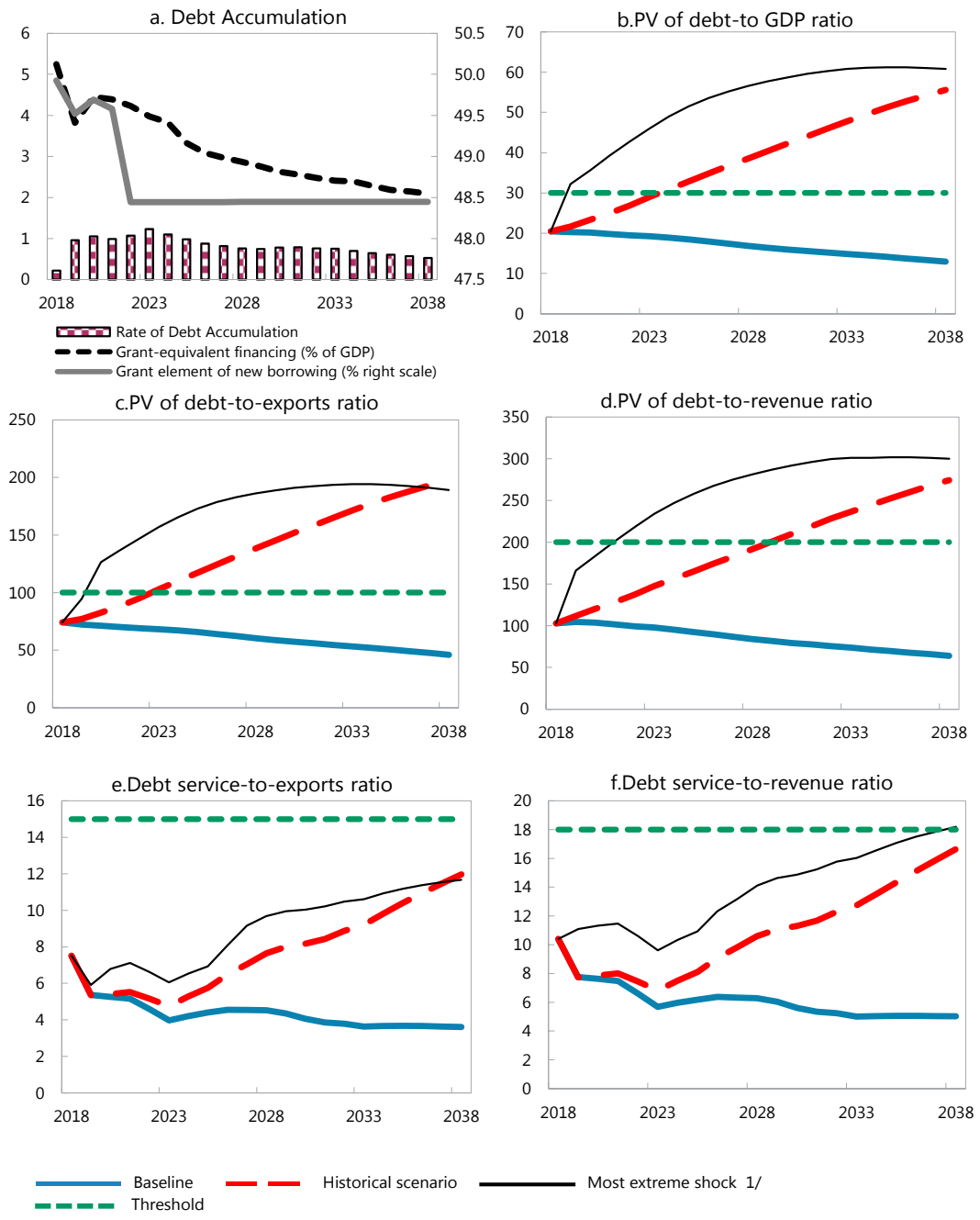
- *Reducing the sovereign-bank nexus*. There is a significant nexus between sovereign debt and the banking system since the RBM and commercial banks have been the main sources of domestic financing for the central government. As highlighted in the Selected Issues Paper,[8] such large

[7] For instance, taking a $600 million loan over the next three years, with a grant element of 37 percent, would raise the PV of debt to GDP ratio by 5 percentage points at the peak and lead the PV of debt to exports ratio to hit the 100 percent threshold.
[8] Selected Issues Paper on "Supporting growth through increased credit to the private sector".

exposure of bank balance sheets to the sovereign risks a negative feedback loop if fiscal challenges emerge or liquidity conditions tighten.

- *Closely monitoring contingent liabilities.* Contingent liabilities have in general been one of the largest sources of fiscal risk. Malawi's experience in the past few years, such as the recapitalization of the RBM due to devaluation and the ADMARC bailout, has illustrated that the materialization of contingent liabilities can contribute to unexpected increases in debt-to-GDP ratio, crowding out private credit and jeopardizing debt sustainability. Efforts should be stepped up to estimate, disclose, manage, and contain contingent liabilities, in particular of those related to financial sector and state-owned enterprises.

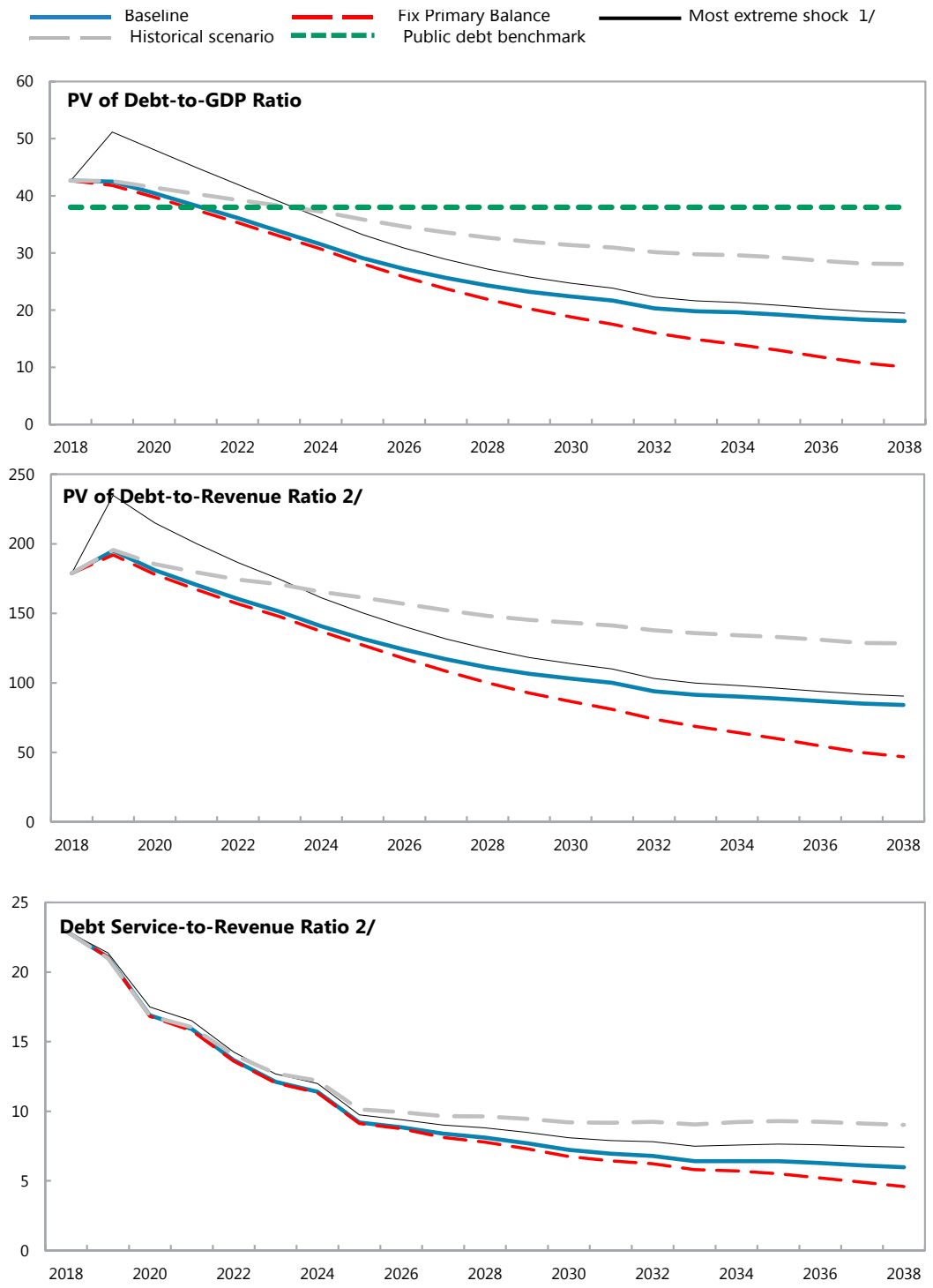
Figure 1. Indicators of Public and Publicly Guaranteed External Debt under Alternatives Scenarios, 2018–38 1/



Sources: Malawian authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio on or before 2028. In figure b. it corresponds to a One-time depreciation shock; in c. to a Exports shock; in d. to a One-time depreciation shock; in e. to a Exports shock and in figure f. to a One-time depreciation shock

Figure 2. Indicators of Public Debt Under Alternative Scenarios, 2018–38 1/



Sources: Malawian authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio on or before 2028.

2/ Revenues are defined inclusive of grants.

Table 1. External Debt Sustainability Framework, Baseline Scenario, 2015–38

(Percent of GDP, unless otherwise indicated)

	Actual			Historical Average 6/	Standard Deviation 6/	Projections									
	2015	2016	2017			2018	2019	2020	2021	2022	2023	2018-2023 Average	2028	2038	2024-2038 Average
External debt (nominal) 1/	**40.5**	**37.3**	**36.3**			**35.6**	**35.6**	**35.6**	**35.3**	**34.7**	**34.1**		**30.2**	**23.8**	
of which: public and publicly guaranteed (PPG)	37.0	33.2	32.6			32.1	32.2	32.2	32.0	31.6	31.2		27.9	22.3	
Change in external debt	3.5	-3.2	-1.0			-0.7	0.1	0.0	-0.4	-0.6	-0.5		-0.8	-0.6	
Identified net debt-creating flows	5.7	17.5	3.6			5.6	4.3	3.8	3.4	3.2	3.1		2.4	1.7	
Non-interest current account deficit	**8.6**	**12.9**	**9.6**	**9.9**	**2.3**	**8.6**	**7.8**	**7.6**	**7.4**	**7.4**	**7.3**		**6.4**	**5.4**	6.1
Deficit in balance of goods and services	11.1	16.8	13.7			13.2	11.3	11.1	11.1	11.1	11.1		10.5	9.6	
Exports	25.5	29.3	27.3			27.5	28.1	28.3	28.3	28.2	28.2		27.9	28.1	
Imports	36.6	46.0	41.0			40.7	39.4	39.4	39.4	39.4	39.3		38.4	37.8	
Net current transfers (negative = inflow)	-4.9	-5.8	-6.0	-5.7	1.4	-6.4	-5.3	-5.4	-5.5	-5.6	-5.6		-5.7	-5.8	-5.7
of which: official	-0.4	0.0	-0.2			-1.2	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Other current account flows (negative = net inflow)	2.4	1.9	1.9			1.9	1.9	1.9	1.8	1.8	1.8		1.7	1.5	
Net FDI (negative = inflow)	**-1.8**	**-3.1**	**-2.0**	**-1.9**	**1.1**	**-2.2**	**-2.3**	**-2.4**	**-2.5**	**-2.5**	**-2.5**		**-2.5**	**-2.5**	-2.5
Endogenous debt dynamics 2/	**-1.2**	**7.7**	**-4.0**			**-0.9**	**-1.2**	**-1.4**	**-1.5**	**-1.7**	**-1.8**		**-1.6**	**-1.2**	
Contribution from nominal interest rate	0.8	0.8	0.4			0.3	0.3	0.3	0.3	0.3	0.3		0.3	0.3	
Contribution from real GDP growth	-1.0	-1.1	-1.3			-1.2	-1.5	-1.7	-1.8	-2.0	-2.1		-1.9	-1.5	
Contribution from price and exchange rate changes	-1.0	7.9	-3.1			...	...	...	...	...	...		...	...	
Residual (3-4) 3/	**1.9**	**-15.0**	**3.4**			**-1.5**	**0.7**	**0.7**	**1.4**	**1.4**	**1.5**		**1.3**	**1.4**	
of which: exceptional financing	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
PV of external debt 4/	...	...	24.8			23.9	23.8	23.5	23.1	22.6	22.2		19.2	14.5	
In percent of exports	...	...	90.8			87.0	84.6	83.2	81.5	80.0	78.9		68.9	51.4	
PV of PPG external debt	**...**	**...**	**21.1**			**20.4**	**20.3**	**20.1**	**19.8**	**19.5**	**19.2**		**16.9**	**12.9**	
In percent of exports	**...**	**...**	**77.2**			**74.2**	**72.2**	**71.3**	**70.0**	**69.0**	**68.3**		**60.5**	**46.0**	
In percent of government revenues	**...**	**...**	**107.2**			**102.7**	**104.5**	**103.7**	**101.6**	**99.4**	**97.9**		**83.9**	**63.9**	
Debt service-to-exports ratio (in percent)	**7.4**	**12.3**	**10.4**			**7.5**	**5.4**	**5.3**	**5.2**	**4.6**	**4.0**		**4.5**	**3.6**	
PPG debt service-to-exports ratio (in percent)	**7.4**	**12.3**	**10.4**			**7.5**	**5.4**	**5.3**	**5.2**	**4.6**	**4.0**		**4.5**	**3.6**	
PPG debt service-to-revenue ratio (in percent)	**10.2**	**19.3**	**14.4**			**10.4**	**7.8**	**7.6**	**7.5**	**6.6**	**5.7**		**6.3**	**5.0**	
Total gross financing need (Billions of U.S. dollars)	0.6	0.8	0.7			0.6	0.5	0.5	0.5	0.5	0.6		0.7	1.1	
Non-interest current account deficit that stabilizes debt ratio	5.1	16.1	10.6			9.3	7.8	7.6	7.8	7.9	7.8		7.3	6.0	
Key macroeconomic assumptions															
Real GDP growth (in percent)	2.9	2.3	4.0	5.0	2.2	3.5	4.5	5.0	5.5	6.0	6.5	5.2	6.5	6.5	6.5
GDP deflator in US dollar terms (change in percent)	2.7	-16.4	9.0	-0.6	13.2	5.0	-0.4	0.8	0.7	0.9	1.2	1.4	1.2	0.6	1.0
Effective interest rate (percent) 5/	2.3	1.8	1.3	1.0	0.6	0.9	0.9	0.9	1.0	1.0	1.0	1.0	1.1	1.1	1.1
Growth of exports of G&S (US dollar terms, in percent)	-7.2	-2.0	5.8	7.1	10.9	9.6	6.3	6.4	6.4	6.8	7.4	7.1	7.5	7.5	7.5
Growth of imports of G&S (US dollar terms, in percent)	-2.4	7.4	1.0	6.6	15.3	7.9	0.8	5.7	6.3	6.9	7.4	5.8	7.2	7.4	7.2
Grant element of new public sector borrowing (in percent)	...	...	...	...	...	49.9	49.5	49.7	49.6	48.4	48.4	49.3	48.4	48.4	48.4
Government revenues (excluding grants, in percent of GDP)	18.5	18.6	19.7			19.9	19.4	19.4	19.5	19.6	19.6		20.1	20.2	20.1
Aid flows (in Billions of US dollars) 7/	0.4	0.3	0.5			0.4	0.3	0.4	0.4	0.4	0.4		0.4	0.6	
of which: Grants	0.2	0.1	0.2			0.3	0.2	0.2	0.2	0.2	0.2		0.2	0.3	
of which: Concessional loans	0.1	0.1	0.3			0.1	0.1	0.2	0.1	0.1	0.2		0.2	0.3	
Grant-equivalent financing (in percent of GDP) 8/	...	...	...			5.2	3.8	4.5	4.4	4.2	4.0		2.9	2.1	2.7
Grant-equivalent financing (in percent of external financing) 8/	...	...	...			80.2	72.0	74.2	75.0	75.3	74.4		71.0	70.1	70.9
Memorandum items:															
Nominal GDP (Billions of US dollars)	6.4	5.5	6.2			6.7	7.0	7.4	7.9	8.4	9.1		13.2	27.0	
Nominal dollar GDP growth	5.7	-14.5	13.3			8.7	4.1	5.8	6.2	7.0	7.8	6.6	7.7	7.2	7.5
PV of PPG external debt (in Billions of US dollars)			1.3			1.3	1.4	1.5	1.5	1.6	1.7		2.2	3.5	
(PVt-PVt-1)/GDPt-1 (in percent)						0.2	1.0	1.1	1.0	1.1	1.2	0.9	0.8	0.5	0.8
Gross workers' remittances (Billions of US dollars)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
PV of PPG external debt (in percent of GDP + remittances)	...	...	21.1			20.4	20.3	20.1	19.8	19.5	19.2		16.9	12.9	
PV of PPG external debt (in percent of exports + remittances)	...	...	77.2			74.2	72.2	71.3	70.0	69.0	68.3		60.5	46.0	
Debt service of PPG external debt (in percent of exports + remittances)	...	...	10.4			7.5	5.4	5.3	5.2	4.6	4.0		4.5	3.6	

Sources: Malawian authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.

2/ Derived as [r - g - ρ(1+g)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.

3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.

4/ Assumes that PV of private sector debt is equivalent to its face value.

5/ Current-year interest payments divided by previous period debt stock.

6/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

7/ Defined as grants, concessional loans, and debt relief.

8/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

Table 2. Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2018–38

	Projections							
	2018	2019	2020	2021	2022	2023	**2028**	2038
PV of debt-to GDP ratio								
Baseline	20	20	20	20	19	19	**17**	13
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2018-2038 1/	20	22	23	25	27	29	**39**	56
A2. New public sector loans on less favorable terms in 2018-2038 2/	20	21	21	22	22	23	**23**	22
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2019-2020	20	23	26	29	31	33	**41**	44
B2. Export value growth at historical average minus one standard deviation in 2019-2020 3/	20	24	29	32	34	36	**42**	44
B3. US dollar GDP deflator at historical average minus one standard deviation in 2019-2020	20	26	34	37	40	43	**53**	57
B4. Net non-debt creating flows at historical average minus one standard deviation in 2019-2020 4/	20	24	28	30	33	35	**41**	43
B5. Combination of B1-B4 using one-half standard deviation shocks	20	26	35	38	40	43	**50**	52
B6. One-time 30 percent nominal depreciation relative to the baseline in 2019 5/	20	32	36	39	43	46	**57**	61
PV of debt-to-exports ratio								
Baseline	74	72	71	70	69	68	**61**	46
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2018-2038 1/	74	77	83	89	95	103	**138**	197
A2. New public sector loans on less favorable terms in 2018-2038 2/	74	73	75	77	79	81	**83**	79
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2019-2020	74	80	88	97	106	114	**142**	151
B2. Export value growth at historical average minus one standard deviation in 2019-2020 3/	74	94	127	137	147	157	**186**	189
B3. US dollar GDP deflator at historical average minus one standard deviation in 2019-2020	74	80	88	97	106	114	**142**	151
B4. Net non-debt creating flows at historical average minus one standard deviation in 2019-2020 4/	74	85	98	107	115	124	**149**	153
B5. Combination of B1-B4 using one-half standard deviation shocks	74	90	113	123	132	141	**167**	169
B6. One-time 30 percent nominal depreciation relative to the baseline in 2019 5/	74	80	88	97	106	114	**142**	151
PV of debt-to-revenue ratio								
Baseline	103	105	104	102	99	98	**84**	64
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2018-2038 1/	103	111	120	128	137	147	**192**	274
A2. New public sector loans on less favorable terms in 2018-2038 2/	103	106	110	112	114	116	**115**	110
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2019-2020	103	118	133	146	158	170	**204**	218
B2. Export value growth at historical average minus one standard deviation in 2019-2020 3/	103	124	151	163	174	184	**211**	215
B3. US dollar GDP deflator at historical average minus one standard deviation in 2019-2020	103	134	173	190	206	221	**266**	283
B4. Net non-debt creating flows at historical average minus one standard deviation in 2019-2020 4/	103	123	143	155	166	177	**206**	213
B5. Combination of B1-B4 using one-half standard deviation shocks	103	134	179	193	206	218	**251**	255
B6. One-time 30 percent nominal depreciation relative to the baseline in 2019 5/	103	166	184	201	218	234	**281**	300

Table 3. Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2018–38 (cont.)

Debt service-to-exports ratio								
Baseline	8	5	5	5	5	4	**5**	4
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2018-2038 1/	8	5	5	6	5	5	**8**	12
A2. New public sector loans on less favorable terms in 2018-2038 2/	8	5	5	5	5	5	**5**	5
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2019-2020	8	5	5	6	5	5	**7**	9
B2. Export value growth at historical average minus one standard deviation in 2019-2020 3/	8	6	7	7	7	6	**10**	12
B3. US dollar GDP deflator at historical average minus one standard deviation in 2019-2020	8	5	5	6	5	5	**7**	9
B4. Net non-debt creating flows at historical average minus one standard deviation in 2019-2020 4/	8	5	6	6	5	5	**8**	9
B5. Combination of B1-B4 using one-half standard deviation shocks	8	6	6	6	6	5	**9**	10
B6. One-time 30 percent nominal depreciation relative to the baseline in 2019 5/	8	5	5	6	5	5	**7**	9
Debt service-to-revenue ratio								
Baseline	10	8	8	7	7	6	**6**	5
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2018-2038 1/	10	8	8	8	7	7	**11**	17
A2. New public sector loans on less favorable terms in 2018-2038 2/	10	8	8	8	7	7	**7**	8
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2019-2020	10	8	8	8	8	7	**10**	13
B2. Export value growth at historical average minus one standard deviation in 2019-2020 3/	10	8	8	8	8	7	**11**	13
B3. US dollar GDP deflator at historical average minus one standard deviation in 2019-2020	10	9	11	11	10	9	**13**	17
B4. Net non-debt creating flows at historical average minus one standard deviation in 2019-2020 4/	10	8	8	8	8	7	**11**	13
B5. Combination of B1-B4 using one-half standard deviation shocks	10	8	10	10	9	8	**13**	16
B6. One-time 30 percent nominal depreciation relative to the baseline in 2019 5/	10	11	11	11	11	10	**14**	18
Memorandum item:								
Grant element assumed on residual financing (i.e., financing required above baseline) 6/	44	44	44	44	44	44	**44**	44

Sources: Malawian authorities; and staff estimates and projections.

1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.

2/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline, while grace and maturity periods are the same as in the baseline.

3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly an offsetting adjustment in import levels).

4/ Includes official and private transfers and FDI.

5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.

6/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.

Table 4. Public Sector Debt Sustainability Framework, Baseline Scenario, 2015–38

(Percent of GDP, unless otherwise indicated)

	Actual					Estimate	Projections								
	2015	2016	2017	Average 5/	Standard Deviation 5/	2018	2019	2020	2021	2022	2023	2018-23 Average	2028	2038	2024-38 Average
Public sector debt 1/	54.4	54.4	55.1			54.3	54.4	52.6	50.5	48.2	45.7		35.3	27.5	
of which: foreign-currency denominated	37.0	33.2	32.6			32.1	32.2	32.2	32.0	31.6	31.2		27.9	22.3	
Change in public sector debt	6.2	0.0	0.7			-0.8	0.1	-1.8	-2.1	-2.3	-2.5		-1.5	-0.4	
Identified debt-creating flows	7.6	-0.6	-0.2			-0.6	0.3	-1.9	-1.9	-2.1	-2.5		-1.3	-0.5	
Primary deficit	2.1	2.1	2.9	1.3	1.7	-0.4	0.9	-0.3	-0.3	-0.2	-0.4	-0.1	0.7	1.0	0.7
Revenue and grants	21.7	21.3	23.1			23.9	21.8	22.4	22.5	22.5	22.3		21.9	21.5	
of which: grants	3.2	2.7	3.4			4.0	2.4	2.9	3.0	2.9	2.7		1.8	1.3	
Primary (noninterest) expenditure	23.9	23.4	26.0			23.5	22.6	22.0	22.2	22.4	21.9		22.6	22.5	
Automatic debt dynamics	5.5	-2.4	-3.1			-0.1	-0.6	-1.5	-1.6	-1.9	-2.1		-2.1	-1.5	
Contribution from interest rate/growth differential	-0.4	0.3	-0.5			-0.6	-1.0	-1.9	-1.9	-2.1	-2.3		-2.3	-1.8	
of which: contribution from average real interest rate	1.0	1.5	1.6			1.3	1.3	0.7	0.8	0.7	0.7		0.0	-0.1	
of which: contribution from real GDP growth	-1.4	-1.2	-2.1			-1.9	-2.3	-2.6	-2.7	-2.9	-2.9		-2.2	-1.7	
Contribution from real exchange rate depreciation	5.9	-2.8	-2.6			0.4	0.4	0.3	0.3	0.2	0.2		...	...	
Other identified debt-creating flows	0.0	-0.3	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Privatization receipts (negative)	0.0	-0.3	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Recognition of implicit or contingent liabilities	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Debt relief (HIPC and other)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Other (specify, e.g. bank recapitalization)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Residual, including asset changes	-1.4	0.6	0.9			-0.3	-0.2	0.1	-0.2	-0.2	0.0		-0.2	0.2	
Other Sustainability Indicators															
PV of public sector debt	...	...	43.6			42.7	42.5	40.5	38.3	36.1	33.8		24.3	18.1	
of which: foreign-currency denominated	...	...	21.1			20.4	20.3	20.1	19.8	19.5	19.2		16.9	12.9	
of which: external	...	...	21.1			20.4	20.3	20.1	19.8	19.5	19.2		16.9	12.9	
PV of contingent liabilities (not included in public sector debt)	...	...	...			...	...	...	...	...	...		...	...	
Gross financing need 2/	7.2	9.7	9.4			5.1	5.4	3.4	3.3	2.9	2.3		2.5	2.2	
PV of public sector debt-to-revenue and grants ratio (in percent)	...	...	188.8			178.8	195.1	181.1	170.5	160.5	151.3		111.2	84.2	
PV of public sector debt-to-revenue ratio (in percent)	...	...	221.9			214.4	218.9	208.3	196.3	184.4	172.0		120.9	89.5	
of which: external 3/	...	...	107.2			102.7	104.5	103.7	101.6	99.4	97.9		83.9	63.9	
Debt service-to-revenue and grants ratio (in percent) 4/	22.3	35.8	27.9			22.9	21.1	16.9	15.9	13.7	12.1		8.1	6.0	
Debt service-to-revenue ratio (in percent) 4/	26.1	40.9	32.8			27.5	23.6	19.4	18.3	15.7	13.8		8.8	6.3	
Primary deficit that stabilizes the debt-to-GDP ratio	-4.1	2.1	2.2			0.4	0.8	1.5	1.8	2.1	2.1		2.2	1.3	
Key macroeconomic and fiscal assumptions															
Real GDP growth (in percent)	2.9	2.3	4.0	5.0	2.2	3.5	4.5	5.0	5.5	6.0	6.5	5.2	6.5	6.5	6.5
Average nominal interest rate on forex debt (in percent)	2.6	2.0	1.5	1.2	0.7	1.0	1.0	1.0	1.1	1.1	1.1	1.1	1.2	1.2	1.2
Average real interest rate on domestic debt (in percent)	3.1	7.4	8.0	3.0	4.2	7.6	8.2	5.1	6.0	5.8	5.9	6.4	2.4	2.0	3.1
Real exchange rate depreciation (in percent, + indicates depreciation	17.9	-7.6	-8.2	5.2	26.0	1.4	...	...	...	...	...	...	...	...	...
Inflation rate (GDP deflator, in percent)	20.9	19.5	10.7	16.3	5.9	8.8	6.8	6.1	5.2	4.6	4.4	6.0	4.4	3.8	4.2
Growth of real primary spending (deflated by GDP deflator, in percer	14.0	0.2	15.9	3.0	6.3	-6.8	0.9	2.2	6.1	6.9	4.5	2.3	6.9	4.8	6.7
Grant element of new external borrowing (in percent)	...	...	...	...	...	49.9	49.5	49.7	49.6	48.4	48.4	49.3	48.4	48.4	...

Sources: Malawian authorities; and staff estimates and projections.

1/ Public sector covers central government. Gross debt is used.

2/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.

3/ Revenues excluding grants.

4/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.

5/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

Table 5. Sensitivity Analysis for Key Indicators of Public Debt, 2018–38

	Projections							
	2018	2019	2020	2021	2022	2023	2028	2038
PV of Debt-to-GDP Ratio								
Baseline	43	43	40	38	36	34	24	18
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	43	43	41	40	39	38	33	28
A2. Primary balance is unchanged from 2018	43	42	40	38	35	33	22	10
A3. Permanently lower GDP growth 1/	43	43	41	39	37	35	28	29
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2019-2020	43	43	43	41	39	37	29	25
B2. Primary balance is at historical average minus one standard deviations in 2019-2020	43	44	43	41	39	37	27	19
B3. Combination of B1-B2 using one half standard deviation shocks	43	43	43	41	39	37	28	22
B4. One-time 30 percent real depreciation in 2019	43	51	48	45	42	39	27	19
B5. 10 percent of GDP increase in other debt-creating flows in 2019	43	48	46	43	41	39	28	20
PV of Debt-to-Revenue Ratio 2/								
Baseline	179	195	181	170	160	151	111	84
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	179	196	186	180	174	171	148	129
A2. Primary balance is unchanged from 2018	179	192	178	167	157	148	100	47
A3. Permanently lower GDP growth 1/	179	196	183	174	165	158	127	133
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2019-2020	179	199	190	181	172	164	131	115
B2. Primary balance is at historical average minus one standard deviations in 2019-2020	179	201	194	183	173	163	121	90
B3. Combination of B1-B2 using one half standard deviation shocks	179	199	193	183	173	164	126	101
B4. One-time 30 percent real depreciation in 2019	179	235	215	200	187	175	124	91
B5. 10 percent of GDP increase in other debt-creating flows in 2019	179	220	205	193	183	173	129	95
Debt Service-to-Revenue Ratio 2/								
Baseline	23	21	17	16	14	12	8	6
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	23	21	17	16	14	13	10	9
A2. Primary balance is unchanged from 2018	23	21	17	16	14	12	8	5
A3. Permanently lower GDP growth 1/	23	21	17	16	14	12	9	8
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2019-2020	23	21	17	17	14	13	9	7
B2. Primary balance is at historical average minus one standard deviations in 2019-2020	23	21	17	16	14	12	9	6
B3. Combination of B1-B2 using one half standard deviation shocks	23	21	17	16	14	13	9	7
B4. One-time 30 percent real depreciation in 2019	23	23	20	19	16	15	11	9
B5. 10 percent of GDP increase in other debt-creating flows in 2019	23	21	17	17	14	13	9	7

Sources: Malawian authorities; and staff estimates and projections.
1/ Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of the length of the projection period.
2/ Revenues are defined inclusive of grants.